UNDER REVIEW

RESCIENCE C

Replication

[RE] A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment

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A replication of *"A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment"* from Hou et al. (2021), published in the *2021 IEEE Intelligent Transportation Systems Conference (ITSC)*, Indianapolis, USA, September 19–21, 2021.

Replication Summary

Scope of Replication — The objective of this work is to reproduce the results of the reference study [1] and verify the claim that the reinforcement learning algorithms Ape-X DQN, PPO and A3C show superior performance for freeway ramp meter signal control in the metrics traffic throughput, vehicle speed, number of stops per vehicle, fuel efficiency per vehicle and per-vehicle emission rates for CO_2 and NO_x than the given baseline controllers no ramp meter, fixed-time controller and ALINEA in both mixed near-congested/congested traffic situations and extremely congested traffic situations. Furthermore, we aim to confirm that PPO shows better results in all of the aforementioned assessment criteria than the other reinforcement learning algorithms, while showing a convergence trend similar to Ape-X DQN and one that is approximately 15 times faster than A3C. To achieve this, we implemented a simulation environment in Python, together with the three baseline controllers (no ramp meter, fixed-time control and ALINEA) and the three RL-based controllers (Ape-X DQN, PPO and A3C), which we subsequently used to compare our obtained results with the original claims.

Methodology — Since the authors of the reproduced study did not provide any source code, the modelling of the environment, the route generation, the network, the reward function, the state computation, the three baseline controllers, the three reinforcement learning algorithms and also the training and evaluation scripts were implemented from scratch based on the paper description. For Ape-X DQN, PPO and A3C we made use of the reinforcement learning library RLlib [2] to remain consistent with the original study and the microscopic traffic simulations were conducted using the SUMO software (Simulation of Urban Mobility) together with TraCI (Traffic Control Interface) [3].

Results — Our replication findings do not match the original work from [1]. With their described parameters, the baseline controllers (no ramp meter, fixed-time ramp meter and ALINEA) already showed notable deviations in the evaluation metrics. Furthermore, the hypothesis that the reinforcement learning policies outperform the baseline

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The authors have declared that no competing interests exists.
Code is available at https://github.com/DerKevinRiehl/replication_ramp_metering_rl_2021_houetal.

controllers could not be reproduced. Moreover, we were not able to provide evidence that PPO surpasses Ape-X DQN and A3C in our measurements. However, we can confirm that PPO converges at a similar rate as Ape-X DQN and that A3C takes significantly longer.

What was easy — The implementation of the three baseline-feedback controllers (no ramp meter, fixed-time controller and ALINEA) was straightforward, especially no ramp meter and fixed-time controller. The reason for this is that the paper provided both formulas and parameter values for ALINEA, while also specifying the cycle length for the fixed-time controller.

What was difficult — Implementing the environment with all its details required significant effort.

1 Introduction

With the increase of urbanization and global population growth, cities are expected to house an additional 2.5 billion people by 2050 [4]. This rising number of people in cities will inevitably result in higher traffic demand in metropolitan areas. Generally, travel demand surpasses road capacity on a regular basis, especially with poorly coordinated traffic signal phases, and the resulting congestion negatively impacts not only a nation's economy, but also the individual's quality of life [5].
Freeway bottlenecks, particularly on-ramp merging areas, represent one of the major causes of congestion in urban transportation networks [5]. To address this issue, ramp metering strategies have been widely adopted to regulate the flow of vehicles entering freeways and maintain stable traffic conditions, where the red-to-green cycles are driven by a control algorithm [6, 7]. Classical approaches that have been introduced in the literature include fixed-time controllers, feedback-based methods such as ALINEA [8], fuzzy-control [6, 7], neural-network-based approaches [9, 10, 11, 12], model-based predictive controllers [13] as well as techniques combining multiple control paradigms [14].
However, freeway traffic systems remain highly dynamic, stochastic and nonlinear, making it difficult for traditional control methods to generalize across varying traffic conditions. In recent years, reinforcement learning (RL) has demonstrated impressive performance in solving complex decision-making problems through direct interaction with an environment. These domains range from playing games [15, 16, 17] to robotics [18, 19] as well as fields like energy systems [20, 21] and intelligent infrastructure management [22, 23]. Encouraged by these successes, RL has increasingly been adopted in mobility applications such as traffic signal systems [24, 25, 26, 27, 28], decision making in autonomous vehicles [29, 30, 31, 32, 33, 34], dynamic speed regulation [35, 36] and adaptive road pricing strategies [37, 38].
Recently, several studies have also investigated the use of RL for freeway ramp metering control [39, 40]. For instance, Rezaee et al. [41] proposed an RL-based approach that outperformed the widely used ALINEA controller.
Building upon these developments, this work is a reproduction study of the paper by Hou et al. [1], in which three different RL algorithms are applied to the problem of freeway ramp metering control and compared to three baseline controllers.

2 Summary of Original Paper

The original work *"A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment"* by Hou et al. [1] investigates the application of deep reinforcement learning to freeway ramp metering control. The

study compares three conventional baseline controllers — no ramp metering, a fixed-time controller, and ALINEA — with three deep RL algorithms, namely Ape-X DQN [42], PPO [43], and A3C [44] on a freeway network consisting of a two-lane mainline and a single-lane merging on-ramp.
The objective of the proposed controllers is to improve mobility and environmental efficiency under different traffic conditions. To evaluate controller performance, the study considers six different metrics that degrade under congestion.

- Operational Efficiency
  - Traffic throughput (vph)
  - Average vehicle speed (m/s)
  - Average number of stops per vehicle
- Environmental Impact
  - Average vehicle fuel efficiency (mpg)
  - Average CO_2 emissions per vehicle (g/mi)
  - Average NO_x emissions per vehicle (mg/mi)

The paper reports that the average performance of the deep RL policies is superior to each of the baseline controllers across all aforementioned metrics in both mixed near-congested/congested and extremely congested traffic scenarios.

3 Implementation

3.1 Simulation Network

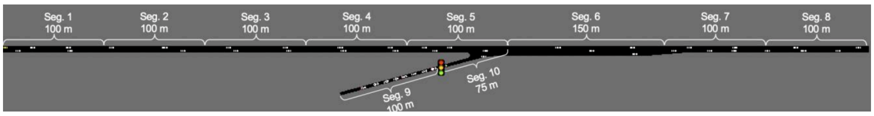

Figure 1. The simulated on-ramp merging area. Figure taken from [1].

The simulation network consists of a two-lane freeway and a single-lane on-ramp merging into the mainline. The freeway is divided into eight segments (Seg. 1 - 8), each with a length of 100 m, except for the merging area (Seg. 6), which has an extended length of 150 m to enable smoother merging.
The on-ramp consists of additional segments, Seg. 9 and Seg. 10, with lengths of 100 m and 75 m, respectively. The ramp meter traffic signal, which is regulated by the implemented control algorithms, is located between Seg. 9 and Seg. 10. At Seg. 6, the network temporarily expands to three lanes as this is where the on-ramp lane gets merged into the freeway, before it tapers down to two lanes again.

3.2 Episode Logic and Traffic Flow Generation

To obtain statistically meaningful results, all controllers were evaluated over 20 simulation episodes (using the microscopic traffic simulator SUMO together with the TraCI interface [3]) and the reported metrics were averaged across all runs. Each episode consisted of 1000 simulation steps. Before the actual evaluation started, an additional warm-up phase of 40 simulation steps was introduced in order to establish realistic traffic conditions before measurements took place.
Two traffic scenarios were considered: mixed near-congested/congested traffic conditions and extremely congested traffic conditions. Following the assumptions made in the original study, the bottleneck capacity of the network using the default SUMO vehicle parameters was estimated to be between 3600 and 3700 veh/h.

In the mixed traffic scenario, the freeway inflow was sampled from a Gaussian distribution $\mathcal{N}(3000, 300)$ veh/h and then clipped to the interval $[0, 3900]$ veh/h. Similarly, the on-ramp inflow was generated independently from $\mathcal{N}(900, 300)$ veh/h and clipped to $[0, 1500]$ veh/h. In the extremely congested traffic scenario, the freeway inflow was fixed at 3900 veh/h, while the on-ramp inflow followed the same distribution as in the mixed traffic scenario, namely $\mathcal{N}(900, 300)$ veh/h clipped to the interval $[0, 1500]$ veh/h. To evaluate the controllers under varying traffic conditions, the traffic inflow on both the freeway and the on-ramp was updated every 100 simulation steps according to the above distributions.

3.3 Emission and Energy Model

For measuring the three metrics average vehicle fuel efficiency (mpg), average CO_2 emissions per vehicle (g/mi) and average NO_x emissions per vehicle (mg/mi), the default emission model in SUMO [45] was used, as specified in the original paper.

3.4 Baseline Controllers

The following baseline controllers were implemented to benchmark the performance of the RL-based approaches.

1. **No Ramp Metering (No RM):** No control is applied, meaning the traffic signal will always be green throughout the entire simulation.

2. **Fixed-Time Control (Fixed):** Ramp metering is controlled by a fixed cycle consisting of a fixed red phase and a fixed green phase. In the original study the total cycle time is 4 seconds, with a 3-second red phase and a 1-second green phase.

3. **ALINEA:** Ramp metering is controlled by a variable cycle consisting of a variable red phase and a 1-second green phase. The cycle time c is recomputed every 15 seconds according to

$$c = \frac{N}{q_{\text{desire}} - q_{\text{freeway}}} \cdot 3600 \quad (1)$$

where:

N is the number of lanes on the ramp (in the given setup $N = 1$),
q_{desire} is the desired total traffic inflow at the merging area bottleneck (vph),
q_{freeway} is the traffic inflow on the freeway.

Moreover, the desired inflow at the bottleneck at time step t is given by

$$q_{\text{desire}}(t) = q_{\text{total}}(t-1) + C_f(n^* - n_{t-1}), \quad (2)$$

where:

$q_{\text{total}}(t-1)$ is the combined inflow of freeway and ramp at the previous time step $t-1$,
C_f is a sensitivity factor,
n^* is the critical number of vehicles at the merging area for maximum traffic outflow,
n_{t-1} is the number of vehicles at the merging area at time step $t-1$.

In the original work, the parameters n^* = 10 and C_f = 20 were used [1].

4 Reinforcement Learning

4.1 State Representation

The state is represented as a 40-dimensional vector with the following three parts:

1. The first 38 entries are derived from the mean speed and number of vehicles on each lane of each segment. As described earlier, the network is split into 10 different segments, hence the dimensionality for each network part is computed as:

$$n_{\text{segments}} \times n_{\text{lanes}} \times n_{\text{variables}},$$

   where $n_{\text{variables}} = 2$, corresponding to the mean speed and the number of vehicles:

$$\overbrace{7 \times 2 \times 2}^{(1)} + \overbrace{2 \times 1 \times 2}^{(2)} + \overbrace{1 \times 3 \times 2}^{(3)} = 38$$

   The first individual term (1) covers segments 1, 2, 3, 4, 5, 7 and 8. The second term (2) addresses the segments on the ramp, namely segments 9 and 10. The third term (3) covers the merging area with three lanes, specifically segment 6.

2. The 39th entry is given by the ramp meter signal (red or green, so 0 or 1 respectively) of the previous simulation step.

3. The 40th and last entry of the state vector depicts the traffic throughput of the merging area bottleneck over the last 30 seconds.

4.2 Action Space

In the considered ramp metering scenario, the traffic signal regulates the vehicle flow from the on-ramp onto the freeway with a two-state signal: red or green. Consequently, the action space is binary.

4.3 Reward Function

While congestion at the merging area (Seg. 6) and before the location of the ramp meter device (Seg. 9) should be heavily penalized, high vehicle speeds in the mentioned network parts should be rewarded, hence the reward function's definition is:

$$R(s_t, a_t) = \frac{\bar{v}_{6,9,10}}{s_{max}} + c_m + c_r \tag{3}$$

where:

- $\bar{v}_{6,9,10}$ is the mean speed on segments 6, 9, and 10.
- s_{max} denotes the maximum speed limit on the freeway.
- c_m is a penalty for congestion in the merging area (Seg. 6), defined as -1 if ≥ 5 vehicles have a speed < 1 m/s.
- c_r represents a penalty for ramp queueing (Seg. 9), set to -1 if ≥ 9 vehicles have a speed < 1 m/s on segment 9.

4.4 Building the RL Agents

To implement and train the agents, the RLlib library was used [2], leveraging 32 CPUs for parallel rollout workers. The neural network architecture remained consistent across all three algorithms, featuring a network with one hidden layer, consisting of 128 neurons and a non-linear activation function, namely tanh. The following parameters were specified in the original work:

Table 1. Hyperparameter settings for the RL algorithms

Parameter	Ape-X DQN	PPO	A3C
Sample size (per iteration)	1,000	32,000	500
Learning rate	5×10^{-4}	5×10^{-5}	1×10^{-4}
Exploration (ϵ)	$0.1 \to 10^{-4}$	–	–
Replay buffer size	5,000,000	–	–
Convergence (steps)	$\sim 20 \times 10^6$	$\sim 20 \times 10^6$	$\sim 300 \times 10^6$

Note: A dash (–) indicates that the parameter was not specified as it is not applicable to the algorithm (e.g., PPO and A3C are on-policy and use neither ϵ-greedy exploration nor a replay buffer).

The original study claims that the reward curve of all three algorithms converged in training. More specifically, the paper reports that the reward of Ape-X DQN and PPO plateaued after about 20 000 000 training steps, whilst it took approximately 300 000 000 training steps for A3C.
Since the original paper does not specify how random seeds are handled across the parallel rollout workers, we make the following assumption. During training, each worker draws a unique seed for every episode, so that no two workers ever encounter the same traffic realisation. A single seed jointly controls the sampled traffic demand and SUMO's internal stochasticity, ensuring that each episode is internally consistent. The motivation behind this scheme is to prevent the policy from overfitting to a particular inflow pattern, which would otherwise be a likely outcome if all workers shared the same seed. For the evaluation runs we instead use the episode index as the seed for both demand generation and SUMO. This yields 20 mutually distinct but fully reproducible evaluation episodes, identical across all evaluated controllers, so that the RL agents and baselines are compared on exactly the same set of traffic realisations.

5 Results: Default Parameters

After implementing the described environment and controllers with all specified parameters, the following results were obtained for the baseline controllers in the mixed near-congested/congested traffic scenario.

5.1 Comparison: Replication vs Paper Baselines

Table 2. Replication vs Paper Values (Mixed Traffic, Default Scenario)

Metric	Replication	Paper	Deviation
ALINEA			
Throughput (veh/h)	3843.66	3664.71	+4.88%
Speed (m/s)	11.87	12.42	−4.43%
Stops per Vehicle	0.86	11.33	−92.41%
Fuel Efficiency (mpg)	19.47	17.00	+14.53%
CO_2 Emissions (g/mi)	479.55	512.99	−6.52%
NO_x Emissions (mg/mi)	194.11	208.00	−6.68%
Fixed			
Throughput (veh/h)	3846.35	3493.46	+10.10%
Speed (m/s)	12.56	14.18	−11.42%
Stops per Vehicle	0.89	3.62	−75.41%
Fuel Efficiency (mpg)	19.12	18.26	+4.71%
CO_2 Emissions (g/mi)	499.82	481.71	+3.76%
NO_x Emissions (mg/mi)	203.32	195.00	+4.27%
No Ramp Meter			
Throughput (veh/h)	3855.16	3664.71	+5.20%
Speed (m/s)	12.49	11.05	+13.03%
Stops per Vehicle	0.69	18.89	−96.35%
Fuel Efficiency (mpg)	20.41	16.01	+27.48%
CO_2 Emissions (g/mi)	454.00	548.61	−17.25%
NO_x Emissions (mg/mi)	182.31	226.09	−19.36%

Mixed Traffic Scenario — With the default parameters, the two metrics we identified as most important, traffic throughput and average vehicle speed, show notable deviations from the values reported in the original paper. The deviation in average vehicle speed of up to 13.03% in the no ramp meter scenario was particularly pronounced and motivated the parameter tuning presented in the next section. For compactness, the remaining plots and tables for the default-parameter setup are provided in the appendix.

6 Results: Tuned Parameters

Since the metrics reported by the original work could not be reproduced with the original parameter setting, we empirically tuned the baseline controllers in order to obtain more accurate values for traffic throughput (veh/h) and average vehicle speed (m/s) in the mixed near-congested/congested traffic scenario.
After extensive experimentation and iterative refinement, the following adjustments were introduced (referred to as the *Tuned Setup* in the remainder of this paper):

- For the vehicle type, the minimum time headway between two consecutive vehicles was increased from the default value of $\tau = 1.00$s to $\tau = 1.04$s. A larger value of τ results in a slightly more cautious car-following behaviour and therefore slightly reduces traffic throughput and speed.
- For the fixed-time controller, the cycle length was extended from 4s to 16s, consisting of a 15s red phase and a 1s green phase. Consequently, fewer vehicles were admitted to the freeway per simulation episode, which restricted the on-ramp inflow and therefore reduced congestion at the merging area, leading to higher average vehicle speeds.

- For the ALINEA controller, all original parameters were kept unchanged. However, instead of deriving the two variables q_{freeway} and q_{ramp} from a single snapshot of the most recent simulation step, both were averaged over a sliding window of ten simulation steps. This reduced measurement noise and yielded a more stable control signal, which in our experiments led to a slight increase in both traffic throughput and average vehicle speed.
- The RL controllers themselves were left unchanged but retrained on the tuned vehicle type with $\tau = 1.04$s.

The obtained values for the baseline controllers are compared to the reported values in the original study in the table below.

6.1 Comparison: Replication vs Paper Baselines

Table 3. Replication vs Paper Values (Mixed Traffic, Tuned Scenario)

Metric	Replication	Paper	Deviation
No Ramp Meter			
Throughput (veh/h)	3840.42	3664.71	+4.79%
Speed (m/s)	11.33	11.05	+2.53%
Stops per Vehicle	0.83	18.89	−95.61%
Fuel Efficiency (mpg)	19.29	16.01	+20.49%
CO_2 Emissions (g/mi)	479.98	548.61	−12.51%
NO_x Emissions (mg/mi)	194.38	226.09	−14.03%
Fixed			
Throughput (veh/h)	3849.59	3493.46	+10.19%
Speed (m/s)	14.06	14.18	−0.85%
Stops per Vehicle	1.17	3.62	−67.68%
Fuel Efficiency (mpg)	19.16	18.26	+4.93%
CO_2 Emissions (g/mi)	572.45	481.71	+18.84%
NO_x Emissions (mg/mi)	236.62	195.00	+21.34%
ALINEA			
Throughput (veh/h)	3851.75	3664.71	+5.10%
Speed (m/s)	12.61	12.42	+1.53%
Stops per Vehicle	1.04	11.33	−90.82%
Fuel Efficiency (mpg)	19.00	17.00	+11.76%
CO_2 Emissions (g/mi)	505.95	512.99	−1.37%
NO_x Emissions (mg/mi)	206.10	208.00	−0.91%

Mixed Traffic Scenario — While the deviation in the traffic throughput remained similar to the default configuration, the deviation in average vehicle speed could be decreased to at most 2.53% across all baseline controllers.
The same comparison for the extreme traffic scenario is shown below.

Table 4. Replication vs Paper Values (Extreme Traffic, Tuned Scenario)

Metric	Replication	Paper	Deviation
ALINEA			
Throughput (veh/h)	4421.60	3790.29	+16.66%
Speed (m/s)	6.87	10.62	−35.31%
Stops per Vehicle	4.75	11.73	−59.51%
Fuel Efficiency (mpg)	11.30	16.83	−32.86%
CO_2 Emissions (g/mi)	874.53	518.52	+68.66%
NO_x Emissions (mg/mi)	374.56	212.66	+76.13%
Fixed			
Throughput (veh/h)	4419.80	3648.60	+21.14%
Speed (m/s)	7.12	9.57	−25.60%
Stops per Vehicle	4.50	7.18	−37.33%
Fuel Efficiency (mpg)	11.43	15.82	−27.75%
CO_2 Emissions (g/mi)	926.45	552.63	+67.64%
NO_x Emissions (mg/mi)	398.49	227.03	+75.52%
No Ramp Meter			
Throughput (veh/h)	4420.52	3683.02	+20.02%
Speed (m/s)	6.17	9.45	−34.71%
Stops per Vehicle	4.90	25.42	−80.72%
Fuel Efficiency (mpg)	11.16	15.25	−26.82%
CO_2 Emissions (g/mi)	881.45	575.34	+53.21%
NO_x Emissions (mg/mi)	377.69	236.62	+59.62%

Extreme Traffic Scenario — Since the parameters were tuned specifically to improve the match in the mixed near-congested/congested traffic scenario, the deviations in the extreme traffic scenario did not improve accordingly. However, for completeness, we still report the obtained values for the baseline controllers in the extreme traffic conditions in the above table.

Having established more accurate baseline values in the mixed traffic scenario, we now turn to the three reinforcement learning based controllers. In line with the original study, all three RL agents were trained exclusively on mixed near-congested and congested traffic conditions. The extremely congested case is used solely for testing the trained policies. We first examine their training convergence before comparing their final performance against the baselines introduced above.

6.2 Convergence Analysis of Reinforcement Learning Controllers

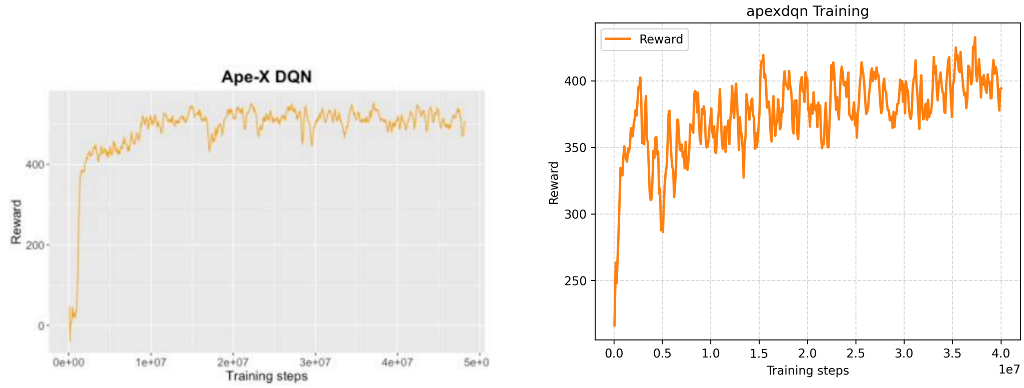

Figure 2. Ape-X DQN training convergence: original (left) vs. replication (right).

Ape-X DQN While the reward curve of Ape-X DQN in the original work shows a clear convergence trend after approximately 20 million training steps, the replicated curve fails to reach a stable convergence trend, ending approximately 100 reward points below the original.

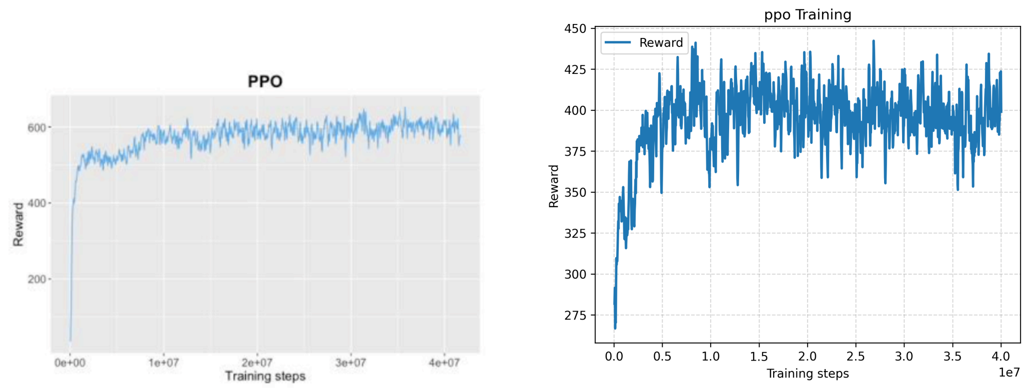

Figure 3. PPO training convergence: original (left) vs. replication (right).

PPO PPO shows a comparable pattern: The replicated reward curve ends roughly 200 points below the value observed in the original reward curve.

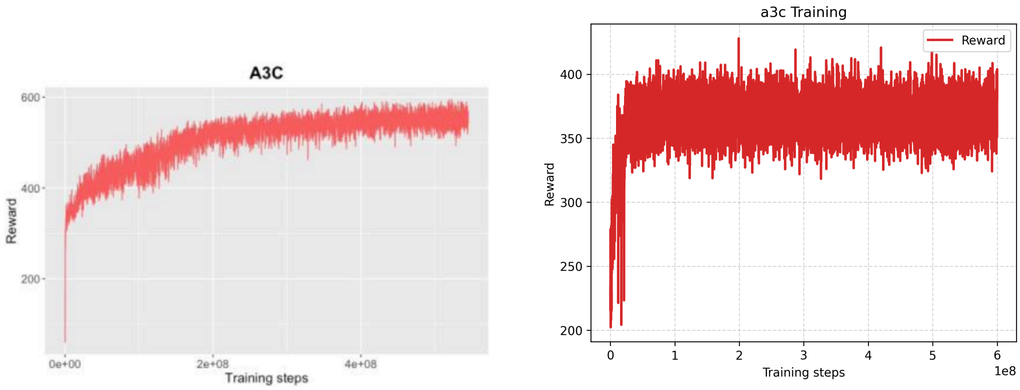

Figure 4. A3C training convergence: original (left) vs. replication (right).

A3C In the original work, A3C converges after 300 million training steps, reaching about 550 reward points at the end of training. In our replication, the reward curve plateaus after around 100 million steps and continues to oscillate between approximately 330–410 reward points. Note that the y-axis of the replication plots is auto-scaled to the value range observed during training and is not anchored at zero, in order to better highlight the shape of the learning curve. The following section evaluates the final performances of the RL controllers and compares them to those of the baseline controllers in a boxplot illustration.

6.3 Boxplots: RL vs Baselines

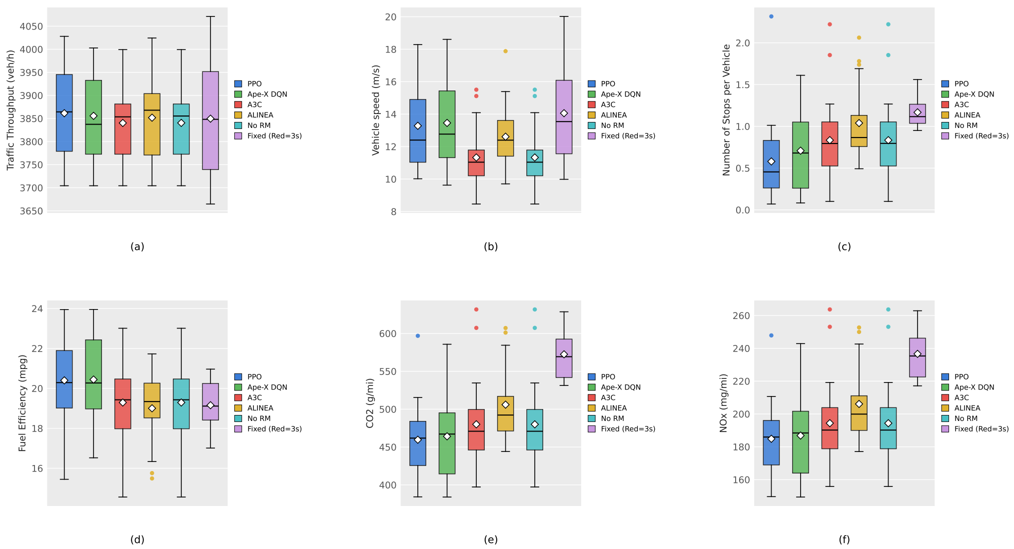

Figure 5. Performance metrics for the mixed traffic scenario. (a) Traffic Throughput (veh/h), (b) Average Vehicle Speed (m/s), (c) Number of Stops per Vehicle, (d) Fuel Efficiency (mpg), (e) CO_2 Emissions (g/mi), (f) NO_x Emissions (mg/mi).

Mixed Traffic Scenario — While Hou et al. report that PPO outperforms the other two RL algorithms in each of the six metrics, our results show that PPO has a slightly lower average vehicle speed and slightly lower fuel efficiency than Ape-X DQN. Therefore, we cannot confirm this claim. Generally, all six controllers show similar performance in traffic throughput, number of stops per vehicle and fuel efficiency. The large performance gap between the RL controllers and the baseline controllers that can be observed in Figure 4 (for example average speeds of around 18–20 m/s, significantly higher fuel efficiency or a lower number of stops for the RL controllers) of the original paper could not be reproduced. Another notable observation is that A3C performs almost identically to the no ramp metering case, suggesting that the learned policy mostly keeps the ramp meter green and switches to red only a few times per episode.

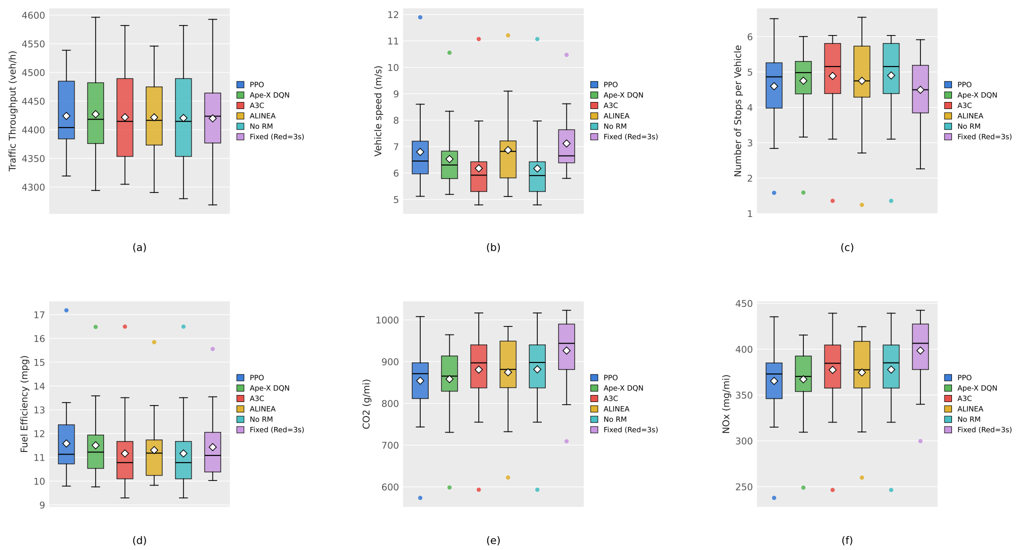

Figure 6. Performance metrics for the extreme traffic scenario. (a) Traffic Throughput (veh/h), (b) Average Vehicle Speed (m/s), (c) Number of Stops per Vehicle, (d) Fuel Efficiency (mpg), (e) CO_2 Emissions (g/mi), (f) NO_x Emissions (mg/mi).

Extreme Traffic Scenario — In the extreme traffic scenario, the three RL-based controllers as well as the baseline controllers result in very similar performance across all six metrics. The means of the throughput are almost identical, whereas in Figure 5 of the original paper PPO and A3C clearly outperform all other controllers in throughput and speed. In our reproduction, no clear winner can be identified and notably the fixed-time controller achieves the highest average vehicle speed. In this scenario, A3C again collapses to a behaviour similar to the no ramp metering case. Even though PPO has the highest fuel efficiency on average and the lowest CO_2 and NO_x emissions, the superiority of PPO in all metrics that is claimed in the paper could not be confirmed.

In the next section, we compare the performance of the RL controller average to the baselines across all metrics and show the percentage deviation of the RL controller average to the baseline values.

6.4 Performance Comparison: RL vs Baselines

Table 5. RL Average vs Baseline Controllers: Mixed Traffic Conditions (Tuned Parameters)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,852.47	+0.02%	+0.31%	+0.07%
Average vehicle speed (m/s)	12.69	+0.63%	+12.00%	−9.74%
Average number of stops per vehicle	0.71	−31.73%	−14.46%	−39.32%
Average vehicle fuel efficiency (mpg)	20.04	+5.47%	+3.89%	+4.59%
Average CO_2 emissions per vehicle (g/mi)	467.92	−7.52%	−2.51%	−18.26%
Average NO_x emissions per vehicle (mg/mi)	188.71	−8.44%	−2.92%	−20.25%

Table 6. RL Average vs Baseline Controllers: Mixed Traffic Conditions (Original Paper)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,738	+2%	+2%	+7%
Average vehicle speed (m/s)	19.0	+53%	+72%	+34%
Average number of stops per vehicle	1.7	−85%	−91%	−53%
Average vehicle fuel efficiency (mpg)	22.1	+30%	+38%	+21%
Average CO_2 emissions per vehicle (g/mi)	395	−23%	−28%	−18%
Average NO_x emissions per vehicle (mg/mi)	156	−25%	−31%	−20%

Mixed Traffic Scenario — The reported superiority of the RL controllers across the given metrics in the table could not be confirmed. For traffic throughput, the RL average is very similar to the baselines, with a maximum increase of 0.31% compared to the no ramp meter controller. Even though the RL average outperforms ALINEA and no ramp metering in terms of average vehicle speed, it is surprisingly about 10% slower than the fixed-time controller. While the RL average clearly outperforms the baseline controllers regarding average number of stops per vehicle, the reported decrease is still significantly higher in the original paper. For the last three metrics (average vehicle fuel efficiency, average CO_2 emissions per vehicle and average NO_x emissions per vehicle), the reproduction results confirm the improvement of the RL average, but not in the magnitude that is reported by Hou et al.

Table 7. RL Average vs Baseline Controllers: Extreme Traffic Conditions (Tuned Parameters)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	4,424.36	+0.06%	+0.09%	+0.10%
Average vehicle speed (m/s)	6.50	−5.39%	+5.35%	−8.71%
Average number of stops per vehicle	4.75	+0.00%	−3.06%	+5.56%
Average vehicle fuel efficiency (mpg)	11.42	+1.06%	+2.33%	−0.09%
Average CO_2 emissions per vehicle (g/mi)	864.45	−1.15%	−1.93%	−6.69%
Average NO_x emissions per vehicle (mg/mi)	369.96	−1.23%	−2.05%	−7.16%

Table 8. RL Average vs Baseline Controllers: Extreme Traffic Conditions (Original Paper)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,904	+3%	+6%	+7%
Average vehicle speed (m/s)	15.4	+45%	+63%	+61%
Average number of stops per vehicle	6.1	−48%	−76%	−15%
Average vehicle fuel efficiency (mpg)	21.2	+26%	+39%	+34%
Average CO_2 emissions per vehicle (g/mi)	420	−19%	−27%	−24%
Average NO_x emissions per vehicle (mg/mi)	168	−21%	−29%	−26%

Extreme Traffic Scenario — For the extreme traffic scenario, the RL average shows very similar performance to the baseline controllers across all six metrics. While traffic throughput is almost identical across all controllers, we could not confirm the reported speed increase of 45–63% for the RL average. In fact, the RL average was even slower than both ALINEA and the fixed-time controller. Furthermore, the paper reports a significant decrease in CO_2 and NO_x emissions, average number of stops per vehicle and a substantial increase in fuel efficiency for the RL average, which we could not confirm in our reproduction. In the last section of the results, we analyse the speed across the freeway. In order to get these plots, we divide the freeway mainline (Seg. 1 - 8, excluding the on-ramp) into 5-meter bins, compute for each bin the mean speed across all simulation steps and evaluation episodes per controller, and render the result using a colormap (0–30 m/s), where brighter colors indicate higher speeds and the red rectangle marks the merging area (Seg. 6).

6.5 Spatial Speed Analysis

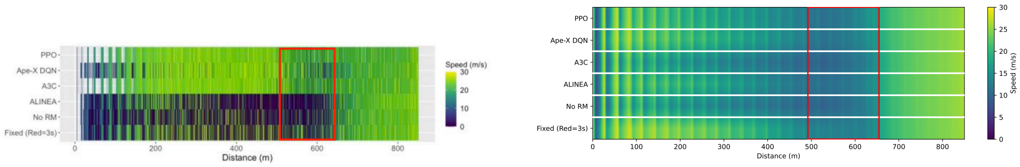

Figure 7. Spatial speed distribution for the mixed traffic scenario: original (left) vs. replication (right).

Mixed Traffic Scenario — While the original paper shows noticeably darker colors in the first 600 m of the freeway for the baseline controllers and predominantly green colors for the RL controllers, indicating a significant speed increase, the reproduced speedmaps display a similar color distribution across all controllers. This suggests that the speed improvement attributed to the RL controllers could not be reproduced.

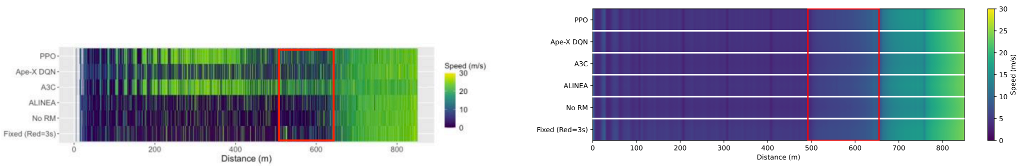

Figure 8. Spatial speed distribution for the extreme traffic scenario: original (left) vs. replication (right).

Extreme Traffic Scenario — Whereas in the original speedmap one can clearly see green colors for the RL controllers, especially PPO and A3C, in the range between 200 m and 600 m, the reproduced speedmap is colored completely blue for all controllers in the said interval. This clearly reveals that the average vehicle speed across the freeway could not be increased by RL-based controllers in the extreme traffic scenario, contrary to the claim of the original paper.

7 Critical Remarks

An initial difficulty in reproducing the study was the lack of a clear specification of how and where the measurements were conducted, which significantly complicated the reproduction process and may explain deviations from the original results. After empirical testing, we adopted the following measurement scheme: traffic throughput was measured locally at Seg. 7 (the first segment downstream of the bottleneck) and the average vehicle speed locally at Seg. 6 (the merging area), while the number of stops per vehicle (with a stop threshold of 1 m/s), fuel efficiency, CO_2 emissions and NO_x emissions were aggregated globally across all vehicles and segments.

Another major challenge in reproducing the work by [1] was identifying the right vehicle parameters and controller settings in order to more closely match the traffic throughput and speed values reported in the original paper, as the proposed default SUMO vehicle parameters did not yield results consistent with those described in their paper.

Additionally, for inflow values sampled from the Gaussian distributions, the generated values were clipped by setting the lower bound to 0. After extensive debugging, it was discovered that SUMO does not allow an inflow value of 0, which caused the episode to terminate immediately and consequently no result file to be written for that episode. Initially, this skewed the measurement results, as only 19 out of 20 episodes actually contributed to the evaluation metrics. This issue could have been considered by the original authors in order to ensure a more robust setup.

Moreover, in the network definition, the lengths of segments ending in a junction are automatically reduced by SUMO, as part of the segment is merged into the junction area. This behaviour was not taken into account by the paper, although additional details regarding whether and how this issue was handled could have been useful for the reproduction of their study.

Furthermore, the claim that only one car can pass the merging area per green phase in the fixed-time controller is not possible to implement, as the traffic signal state cannot be modified during the traci.simulationStep() function in SUMO.

In addition, the training of PPO took significantly longer than reported in the original paper, requiring approximately twice the reported 5 wall-clock hours to reach 20 million training steps. This was observed even though all non-essential computations and logging within the environment were disabled to reduce the computational overhead and save resources, which was not further specified by the original authors. Although 32 CPUs were used for training, as stated in the paper, the difference in training times may stem from the fact that the exact hardware details used by Hou et al. were not disclosed.

The speedmap visualizations in the original paper do not specify the bin size used for aggregating vehicle speeds along the freeway, which makes it difficult to assess the level of detail of the plots and to reproduce them accurately.

Finally, the original paper does not report the absolute performance values of the baseline controllers directly. Instead, they had to be back-calculated from the percentage changes given in Tables I and II, which is unnecessarily cumbersome.

8 Conclusion

This paper presented a replication of the work by Hou et al. [1], who applied reinforcement learning to freeway ramp metering signal control and compared it against three conventional controllers. The replication aimed to verify the claim that the reinforcement learning algorithms Ape-X DQN, PPO and A3C outperform the baseline controllers no ramp metering, fixed-time control and ALINEA under mixed near-congested/congested and extremely congested traffic conditions.

After implementing the environment and controllers as described in the original paper, the obtained results did not match those reported by Hou et al., particularly for the two metrics we consider most important, traffic throughput and average vehicle speed. With a deviation of up to 13.03% in average vehicle speed for the no ramp meter case (mixed traffic scenario), we tuned the simulation parameters empirically to bring the baseline values closer to the reported ones.

With the tuned parameters, the deviation in average vehicle speed was reduced to at most 2.53% and the deviation in traffic throughput to at most 10.19% across the baseline controllers. The remaining metrics, however, still showed notable deviations and the claimed performance gains of the reinforcement learning controllers could not be observed, as all three RL algorithms achieved similar performance to the baselines in both the mixed and the extreme traffic scenario. The reward curves of the RL algorithms also did not converge to the values reported in the original paper. One observation that we can confirm is that the training of A3C was significantly slower than that of Ape-X DQN and PPO.

Because the original paper does not specify the exact location and technical configuration of the measurements, several assumptions had to be made when implementing the evaluation metrics.

Overall, the replication revealed that the results reported by Hou et al. could not be reproduced based on the information available in the original paper. The claimed performance gains of the reinforcement learning controllers over the baseline controllers could not be confirmed. A more complete specification of the simulation environment and the measurement procedure would be necessary to enable a fair and conclusive comparison between reinforcement-learning-based and classical ramp metering strategies.

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A Appendix: Full Results for the Default Parameter Setup

For completeness, all tables and figures for the default parameter setup are provided in this appendix, following the same order as the tuned setup in the main text.

A.1 Comparison: Replication vs Paper Baselines (Extreme Traffic, Default Scenario)

Table 9. Replication vs Paper Values (Extreme Traffic, Default Scenario)

Metric	Replication	Paper	Deviation
ALINEA			
Throughput (veh/h)	4435.08	3790.29	+17.01%
Speed (m/s)	7.01	10.62	−33.99%
Stops per Vehicle	4.47	11.73	−61.89%
Fuel Efficiency (mpg)	11.64	16.83	−30.84%
CO_2 Emissions (g/mi)	836.71	518.52	+61.37%
NO_x Emissions (mg/mi)	357.29	212.66	+68.01%
Fixed			
Throughput (veh/h)	4432.57	3648.60	+21.49%
Speed (m/s)	7.05	9.57	−26.33%
Stops per Vehicle	4.50	7.18	−37.33%
Fuel Efficiency (mpg)	11.53	15.82	−27.12%
CO_2 Emissions (g/mi)	850.49	552.63	+53.90%
NO_x Emissions (mg/mi)	363.57	227.03	+60.14%
No Ramp Meter			
Throughput (veh/h)	4433.47	3683.02	+20.38%
Speed (m/s)	6.89	9.45	−27.09%
Stops per Vehicle	4.71	25.42	−81.47%
Fuel Efficiency (mpg)	11.59	15.25	−24.00%
CO_2 Emissions (g/mi)	841.01	575.34	+46.18%
NO_x Emissions (mg/mi)	359.21	236.62	+51.81%

A.2 Convergence Analysis of Reinforcement Learning Controllers (Default)

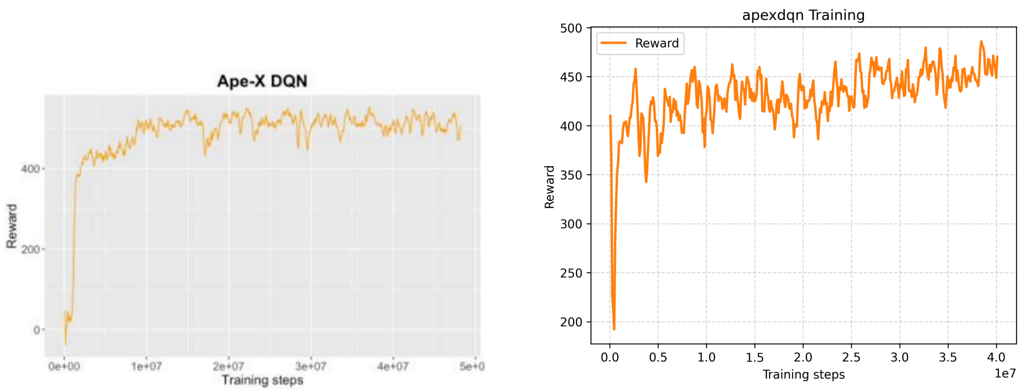

Figure 9. Ape-X DQN training convergence (default setup): original (left) vs. replication (right).

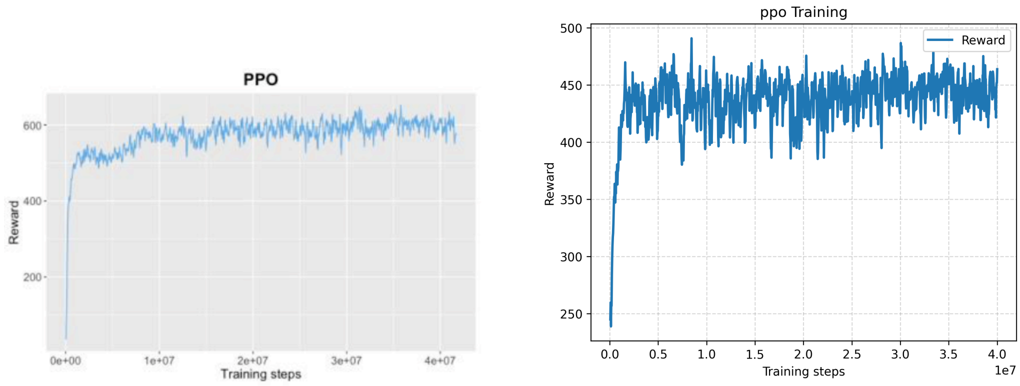

Figure 10. PPO training convergence (default setup): original (left) vs. replication (right).

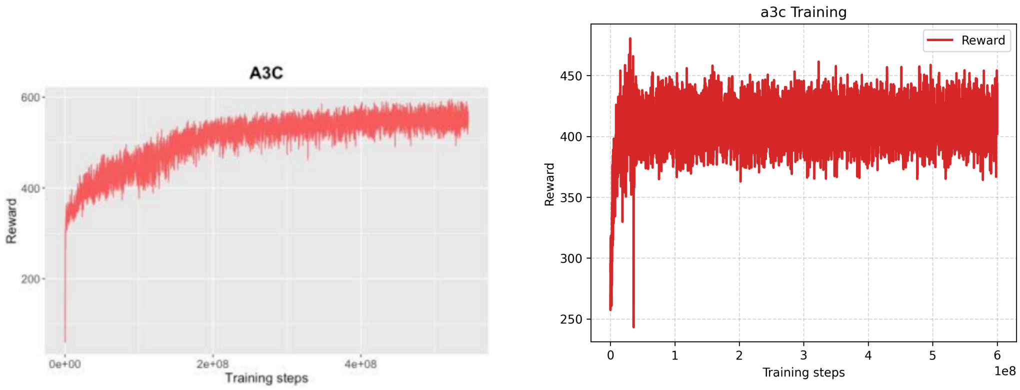

Figure 11. A3C training convergence (default setup): original (left) vs. replication (right).

A.3 Boxplots: RL vs Baselines (Default)

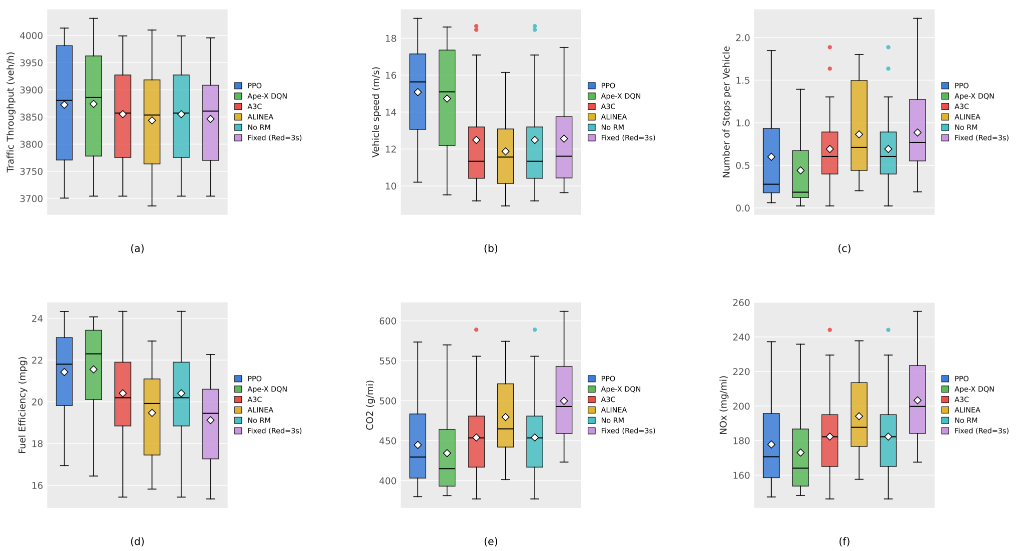

Figure 12. Performance metrics for the mixed traffic scenario (default setup). (a) Traffic Throughput (veh/h), (b) Average Vehicle Speed (m/s), (c) Number of Stops per Vehicle, (d) Fuel Efficiency (mpg), (e) CO_2 Emissions (g/mi), (f) NO_x Emissions (mg/mi).

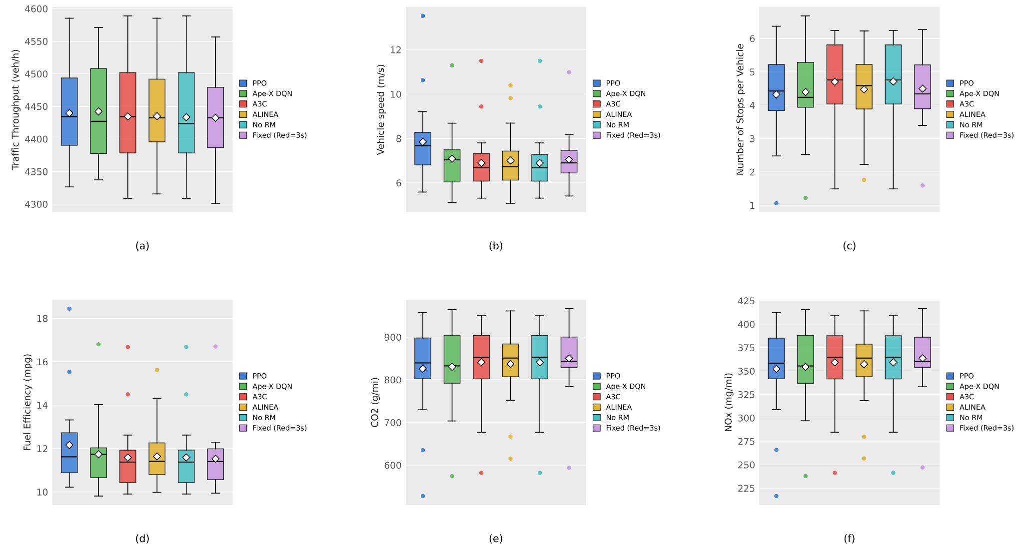

Figure 13. Performance metrics for the extreme traffic scenario (default setup). (a) Traffic Throughput (veh/h), (b) Average Vehicle Speed (m/s), (c) Number of Stops per Vehicle, (d) Fuel Efficiency (mpg), (e) CO_2 Emissions (g/mi), (f) NO_x Emissions (mg/mi).

A.4 Performance Comparison: RL vs Baselines (Default)

Table 10. RL Average vs Baseline Controllers: Mixed Traffic Conditions (Default Parameters)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,867.21	+0.61%	+0.31%	+0.54%
Average vehicle speed (m/s)	14.10	+18.79%	+12.89%	+12.26%
Average number of stops per vehicle	0.58	−32.56%	−15.94%	−34.83%
Average vehicle fuel efficiency (mpg)	21.13	+8.53%	+3.53%	+10.51%
Average CO_2 emissions per vehicle (g/mi)	444.40	−7.33%	−2.11%	−11.09%
Average NO_x emissions per vehicle (mg/mi)	177.73	−8.44%	−2.51%	−12.59%

Table 11. RL Average vs Baseline Controllers: Mixed Traffic Conditions (Original Paper)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,738	+2%	+2%	+7%
Average vehicle speed (m/s)	19.0	+53%	+72%	+34%
Average number of stops per vehicle	1.7	−85%	−91%	−53%
Average vehicle fuel efficiency (mpg)	22.1	+30%	+38%	+21%
Average CO_2 emissions per vehicle (g/mi)	395	−23%	−28%	−18%
Average NO_x emissions per vehicle (mg/mi)	156	−25%	−31%	−20%

Table 12. RL Average vs Baseline Controllers: Extreme Traffic Conditions (Default Parameters)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	4,438.86	+0.09%	+0.12%	+0.14%
Average vehicle speed (m/s)	7.28	+3.85%	+5.66%	+3.26%
Average number of stops per vehicle	4.47	+0.00%	−5.10%	−0.67%
Average vehicle fuel efficiency (mpg)	11.83	+1.63%	+2.07%	+2.60%
Average CO_2 emissions per vehicle (g/mi)	832.31	−0.53%	−1.03%	−2.14%
Average NO_x emissions per vehicle (mg/mi)	355.29	−0.56%	−1.09%	−2.28%

Table 13. RL Average vs Baseline Controllers: Extreme Traffic Conditions (Original Paper)

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,904	+3%	+6%	+7%
Average vehicle speed (m/s)	15.4	+45%	+63%	+61%
Average number of stops per vehicle	6.1	−48%	−76%	−15%
Average vehicle fuel efficiency (mpg)	21.2	+26%	+39%	+34%
Average CO_2 emissions per vehicle (g/mi)	420	−19%	−27%	−24%
Average NO_x emissions per vehicle (mg/mi)	168	−21%	−29%	−26%

A.5 Spatial Speed Analysis (Default)

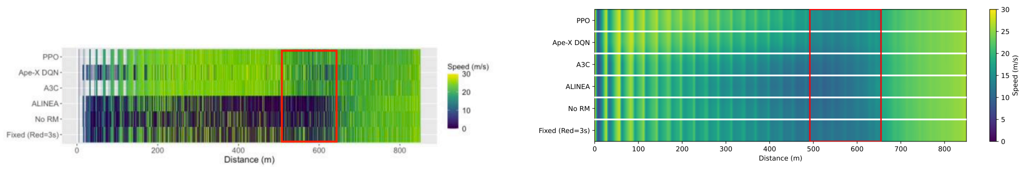

Figure 14. Spatial speed distribution for the mixed traffic scenario (default setup): original (left) vs. replication (right).

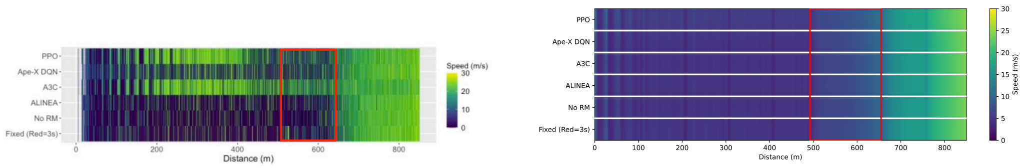

Figure 15. Spatial speed distribution for the extreme traffic scenario (default setup): original (left) vs. replication (right).